

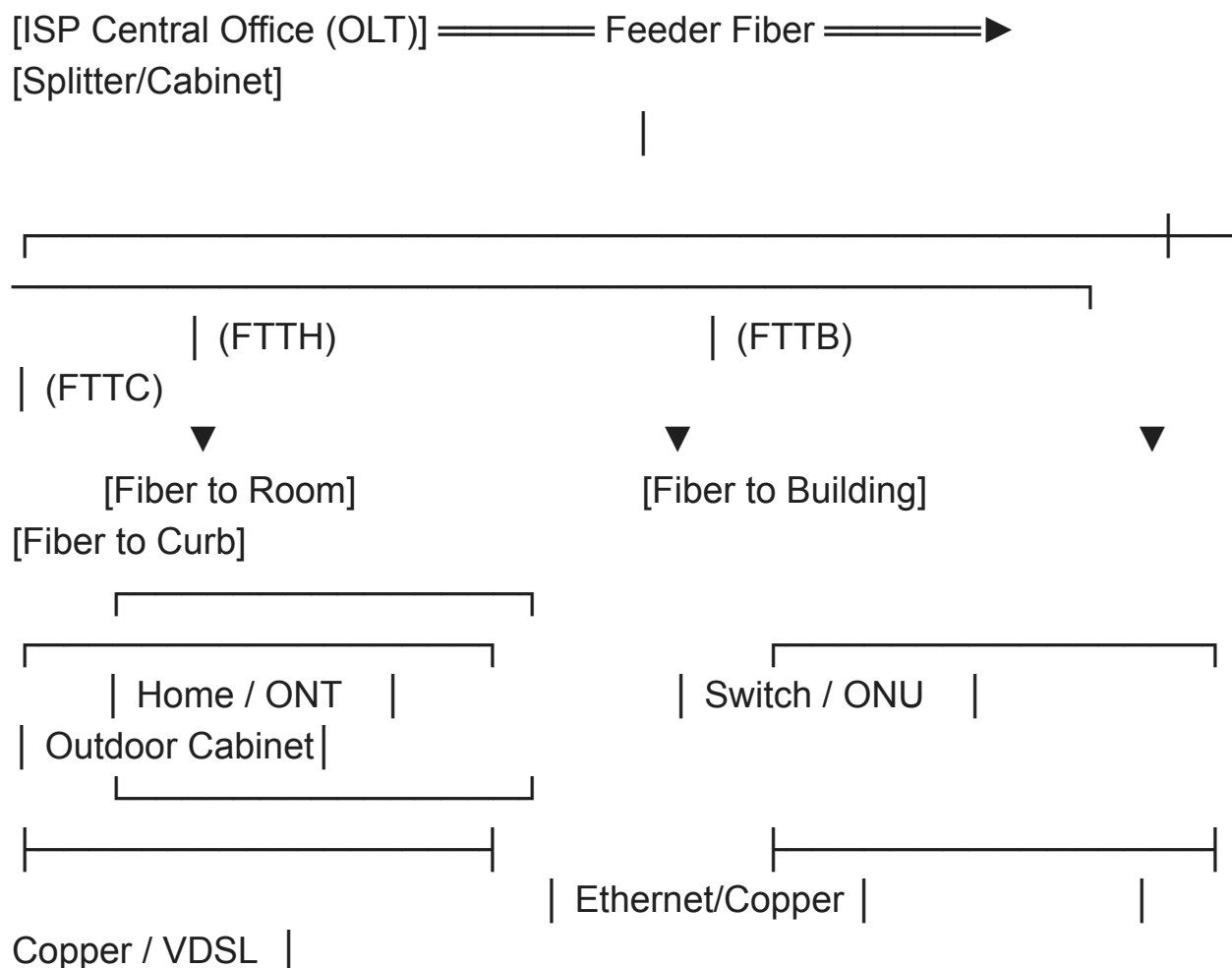
CCNA 200-301: Class Notes — Day 7

Topic: FTTX Architectures, PON Technologies & Satellite vs. Fiber Integration

Modern access networks deliver broadband services to residential, commercial, and remote locations using optical fiber technologies (FTTX) and Satellite internet architectures. Understanding the physical layout, framing standards, passive optical splitting, and deployment hardware is essential for enterprise networking and service provider operations.

1. FTTX Overview & Variations

FTTX (Fiber to the X) is a generic term for any network architecture using optical fiber to replace all or part of the traditional copper local loop.



- **FTTH (Fiber to the Home):** Optical fiber runs directly from the ISP Central Office to individual residences or apartment units, terminating at an **ONT (Optical Network Terminal)**. Offers maximum bandwidth and reliability.
- **FTTB (Fiber to the Building / Business):** Fiber runs to a shared basement or communications room in a multi-dwelling unit (MDU) or enterprise office. Internal distribution uses Ethernet (Cat6) or active switches via an **ONU (Optical Network Unit)**.
- **FTTC (Fiber to the Curb / Cabinet):** Fiber terminates in a street cabinet within a few hundred meters of the subscriber. The final connection to homes uses legacy copper/VDSL.

2. FTTX vs. Starlink (LEO Satellite)

Comparison Vector	FTTX (Fiber Access)	Starlink (LEO Satellite)
Transmission Medium	Glass/Plastic Optical Fiber (Guided Light)	Radio Frequency / Ku & Ka Band to Low Earth Orbit Satellites
Latency	Extremely Low (1 ms – local)	Moderate (25 ms –)
Bandwidth Limits	1 Gbps – symmetric	50 Mbps – shared per cell
SLA & Reliability	High; immune to electromagnetic/weather interference	Susceptible to rain fade, severe storms, and obstructions

Terrain Requirements	Requires physical trenching/cabling right-of-way	Requires clear view of the sky (zero ground infrastructure needed)
Primary Use Case	Urban, suburban, industrial, high-density environments	Remote offshore rigs, isolated islands, disaster relief, moving vessels

3. ISP Optical Network Layering

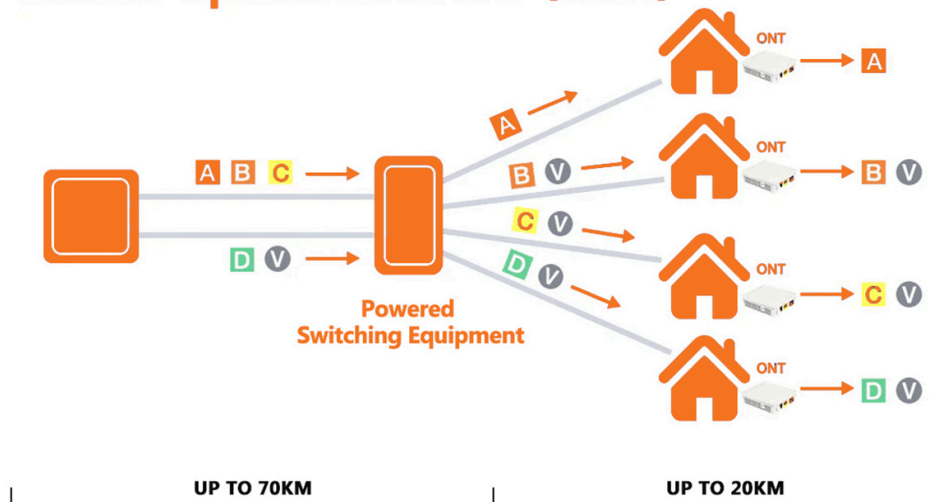
An ISP's optical transport infrastructure is structured into three primary physical segments:

1. **Core Network:** High-speed routed IP/MPLS core connecting Central Offices, IIG gateways, and data centers using high-density DWDM routers.
2. **Feeder Network:** Optical fiber cables spanning from the ISP Central Office (**OLT**) to primary outdoor optical distribution frames (ODF) or passive splitter nodes.
3. **Distribution Network (Last Mile):** Optical cables running from passive splitters to customer drop points and premises terminals.

4. Active (AON) vs. Passive (PON) Optical Networks

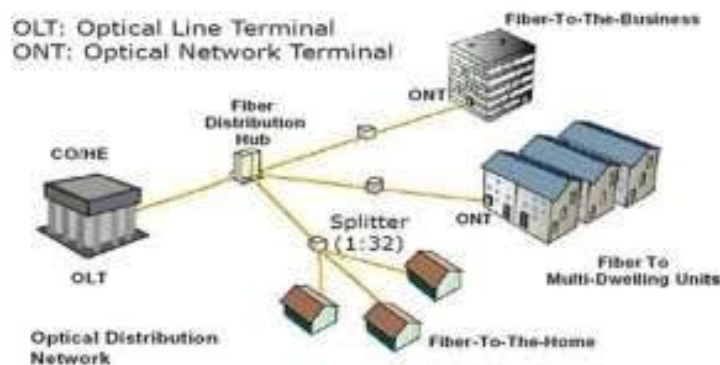
AON (Active Optical Network):

Active Optical Network (AON)



PON (Passive Optical Network):

Passive Optical Network (PON)



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A. AON (Active Optical Network)

- Uses powered electronic switching equipment in the field to direct signals.
- **Pros:** Long transmission distances ($> 100 \text{ km}$), dedicated bandwidth per port.
- **Cons:** High operational expense (requires outdoor electrical power,

backup batteries, active hardware maintenance).

B. PON (Passive Optical Network)

- Uses non-powered optical splitters (glass prisms/couplers) to split a single optical fiber to serve multiple end-users (typically 1:32 or 1:64 ratio).
- **Pros:** Highly cost-effective, zero power required between Central Office and subscriber, simplified maintenance.
- **Cons:** Shared overall bandwidth pool, shorter distance limits (≈ 20 km).

5. PON Standards Evolution

Standard	Full Name	Download Speed	Upload Speed	Primary Framing Mechanism
EPON	Ethernet PON (IEEE 802.3ah)	1.25 Gbps	1.25 Gbps	Native Ethernet Frames
GPON	Gigabit PON (ITU-T G.984)	2.488 Gbps	1.244 Gbps	GEM (GPON Encapsulation Mode) / ATM
XGS-PON	10-Gigabit Symmetric PON	10 Gbps	10 Gbps	Next-Gen GEM Framing
XPON	Dual-Mode Auto-Sensing	Dynamic Support	Dynamic Support	Auto-detects EPON or GPON

	Hardware			signals
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6. Service Bandwidth Models: Dedicated vs. Shared Lines

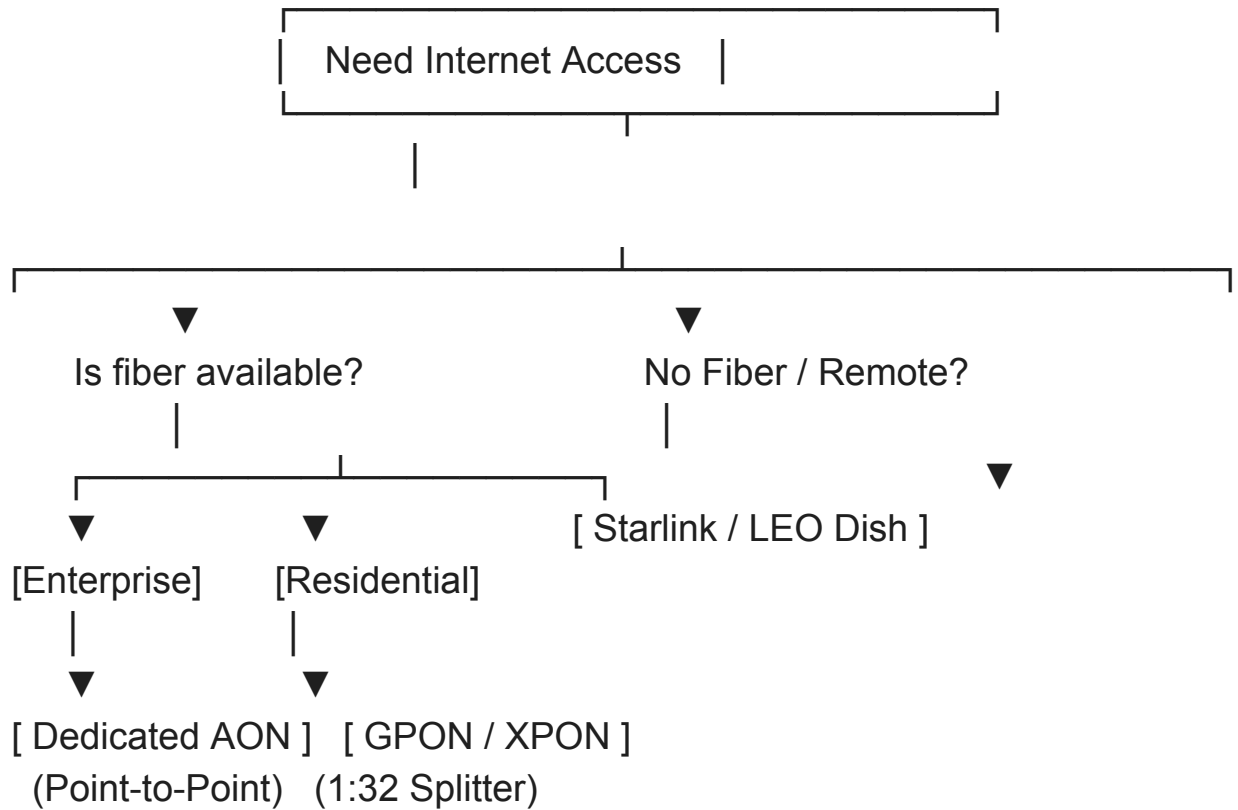
- Shared Line (Broadband Best-Effort):** Multiple subscribers share a single PON port's capacity through Time Division Multiplexing (TDM). Standard for residential internet where users rarely consume max bandwidth simultaneously.
- Dedicated Line (DIA - Dedicated Internet Access):** A dedicated 1:1 fiber connection from the ISP switch directly to the enterprise customer. Guaranteed bandwidth, zero contention ratio, supported by strict Service Level Agreements (SLA) with **99.99%** uptime guarantees.

7. FTTX Hardware Component Breakdown

Hardware Component	Location	Function & Purpose
OLT (Optical Line Terminal)	ISP Central Office	Central master switch/headend that converts IP data from core routers into PON optical signals and manages subscriber bandwidth allocation.
ONU (Optical Network Unit)	Building / Street Cabinet	Medium-scale optical-to-electrical converter serving multiple clients over copper, VDSL, or Ethernet links within a building.

ONT (Optical Network Terminal)	Subscriber Premises	Terminal modem/gateway located inside an end-user's home or office that converts light signals to Ethernet RJ-45 ports / Wi-Fi.
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8. Device Selection Guide & Decision Matrix



1. **When to use Starlink:** Remote vessels, isolated rural clinics, or emergency disaster response where laying optical fiber is physically or economically impossible.
2. **When to use Dedicated AON Lines:** Head offices of commercial

banks, data center interconnects, and mission-critical cloud hosting providers requiring guaranteed 1:1 throughput and strict SLAs.

3. **When to use GPON / XPON (FTTH):** High-density residential housing blocks and suburban home internet delivery where cost per subscriber must be minimized.

9. Real-World Network Design Scenarios

Scenario A: High-Density Residential FTTH Deployment (Uttara, Dhaka)

- **Goal:** Connect 512 residential apartments across 16 residential towers.
- **Design Solution:**
 - **Central Office:** 1x Chassis-Based **OLT** equipped with GPON cards.
 - **Feeder Network:** 8-core optical fiber cable running to street enclosures.
 - **Distribution Network:** 1:32 Passive Optical Splitters installed in basement distribution boxes.
 - **Customer CPE:** Indoor **XPON ONT** devices in each apartment providing Gigabit Ethernet and Wi-Fi 6.

Scenario B: Commercial Bank Head Office & Core Data Center Interconnect

- **Goal:** Connect a primary Data Center in Motijheel to a DR Site in Uttara with zero packet loss and sub-2ms latency.
- **Design Solution:**
 - **Topology:** Point-to-point **AON (Active Optical Network)** Dedicated Fiber Line.
 - **Hardware:** Cisco Catalyst 9500 switches on both ends with 10G/40G LR Transceivers.
 - **Redundancy:** Dual-path fiber routing managed over NTTN backbones (Summit / Fiber@Home) with automatic failover.

Scenario C: Offshore Energy Exploration Rigs in the Bay of Bengal

- **Goal:** Provide real-time data transmission and voice/video connectivity

for engineers stationed off the coast of Cox's Bazar.

- **Design Solution:**

- **Primary Link:** High-Performance **Starlink Business Terminal** mounted on an anti-vibration marine rig mast.
- **Network Edge:** Cisco ISR 1000 series branch router with SD-WAN enabled for encrypted IPSec tunnel transport back to headquarters.